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A hybrid deep learning framework for wind power load forecasting with attention-based feature fusion

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Abstract

This paper presents a hybrid deep learning framework for wind power forecasting that integrates Long Short-Term Memory (LSTM) networks with a Multi-Layer Perceptron (MLP) through an attention-based feature fusion mechanism. The proposed methodology addresses the challenge of simultaneously capturing temporal dependencies in dynamic meteorological sequences and nonlinear mappings of static turbine characteristics. Dynamic features (temperature, pressure, humidity) are processed through LSTM layers, while static features (wind speed/direction at multiple heights) are encoded via MLP with adaptive fusion weights learned through attention. Empirical evaluation on a one-year dataset of 35,040 15-min records from a Xinjiang wind-solar hybrid farm demonstrates that the proposed model achieves a Root Mean Square Error of 26.83 MW and R^2 of 0.71, outperforming direct concatenation (RMSE 28.50 MW, R^2 0.69) and single LSTM (RMSE 52.17 MW, R^2 −0.05). An ablation study further validates that inclusion of photovoltaic parameters as auxiliary atmospheric descriptors reduces RMSE by 2.0% compared to the solar-excluded variant. While Random Forest achieves marginally lower Mean Absolute Error (16.97 vs. 18.18 MW) through effective low-generation regime approximation, the proposed framework offers superior peak-tracking capability (lowest RMSE), balanced error distribution across operational regimes, and scalable modular architecture for evolving sensor infrastructure. The framework provides a reproducible solution for wind power prediction, with particular advantages in handling the intermittent and stochastic nature of wind resources through physics-aligned feature processing and adaptive multi-information fusion.

Keywords Wind power load forecasting, Machine learning, LSTM, Attention mechanism, Feature fusion, Renewable energy integration

1 Introduction

Through the integration of efficient power equipment [1], intelligent monitoring and control system [2], and advanced data analysis and management platform [3], it realizes the automation, intelligent management and optimization of the power system. Intelligent power system has the characteristics of high efficiency and reliability, energy saving,

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safety and environmental protection, and network management, which can significantly improve the operation efficiency, safety and stability of the power system (Fig. 1). As one of the infrastructure of modern society, the stability and efficiency of the electric power system are directly related to the national economic and social development and the quality of people's life [4]. New theoretical directions for power system operation and maintenance and the role of optimization algorithms in the development of sustainable smart grids. The introduction of wisdom theory has improved the quality of people's life to a new height, especially the application of smart power management system. Intelligent power system management refers to the process of real-time monitoring, analysis, optimization and management of intelligent power system with modern information technology [5], automation technology [6] and intelligent control technology [7].

The share of wind power in electricity production continues to increase year after year. In 2023, the total electricity consumption of the whole society in China reached 9223.8 billion kWh, the total power generation in China was 9456.4 billion kWh, up by 6.9% year-on-year. Among this, wind and solar power generation grew rapidly, with a combined increase of 280.1 billion kWh, accounting for 46.1% of the annual increase in total power generation (Fig. 1), accelerating the pace of energy transition to greener sources [8]. Enhancing wind power generation forecasting capabilities can significantly facilitate the green transition. But predictive management in wind power generation faces significant challenges, primarily stemming from the intermittent, stochastic, and uncertain nature of wind resources. Traditional forecasting methods exhibit numerous shortcomings in critical aspects such as data pre-processing, various selection strategies, and typical optimization techniques [9]. Wind energy output is influenced by weather conditions and surrounding topographical environments, with multiple factors—including wind speed, wind direction, temperature, humidity, atmospheric pressure, and altitude—exerting impacts on power generation efficiency [10]. Accurate forecasting of energy consumption and generation is essential for the efficient operation of smart grids, particularly in weather-sensitive environments [11]. Wind power forecasting can use (AI)-based methods [12], the artificial intelligence method now can be adaptive [13] and self-learning in various industries [14], smart grids [15–17], and railway transportation [18], so it is suitable for dealing with the dynamic, nonlinear, and complex characteristics of wind power. Promoting renewable energy utilization by enhancing wind energy forecasting and management to meet the nation's growing electricity demand and provide a viable solution for long-term resource security [19].

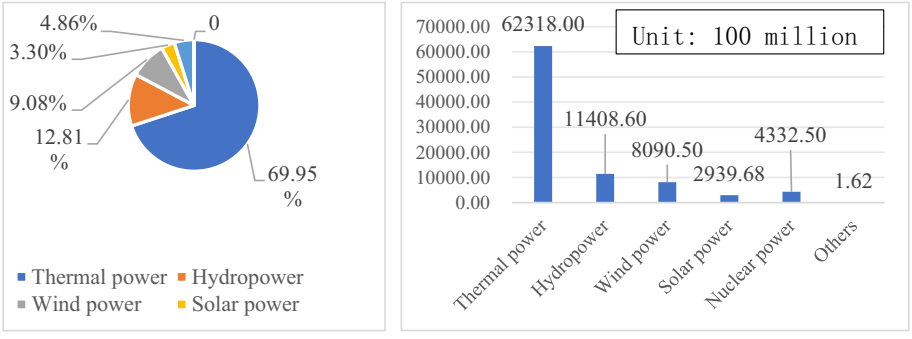


Fig. 1 Share and gross generation of installed power generation capacity by type in 2023

The proposed LSTM-MLP-Attention hybrid deep learning framework specifically addresses this challenge. It utilizes LSTM modules to capture dynamic temporal variations while employing attention fusion layers for weighted optimization of multi-source features, ultimately achieving more accurate and stable wind power load forecasting. While Random Forest achieves the lowest Mean Absolute Error by effectively capturing static feature interactions, it exhibits critical weakness in percentage-based accuracy with MAPE. This limitation arises because tree-based ensemble methods disproportionately accumulate relative errors during low-generation intervals, where small absolute deviations translate to extreme percentage errors. In contrast, the proposed LSTM-MLP-Attention framework achieves superior MAPE performance, representing a 26.6% reduction over Random Forest and a 76.0% reduction over single LSTM. This MAPE superiority stems from the attention mechanism's adaptive weighting of temporal dependencies, which dynamically intensifies model sensitivity during low-generation periods—the precise operational regime where percentage accuracy is most critical for grid reserve scheduling.

However, as the demand for power continues to grow and the power structure becomes increasingly complex, the traditional manual management model struggles to meet the needs for efficient, safe, and reliable operation of modern power systems. Traditional power system management is fraught with limitations that are incompatible with the flourishing of artificial intelligence and intelligent office practices, making it difficult to meet the demands for efficient and precise operations and maintenance in modern power systems. For instance, issues such as low efficiency in manual inspections, lagging fault response, and low data utilization have long constrained the improvement of system reliability. Powered by AI-driven digital and intelligent operations and maintenance technology, a shift from "passive response" to "active early warning" is being achieved. Through deep learning and big data analysis, potential equipment defects can be accurately identified. The vast amount of real-time data generated by the power system, including load data, equipment operation data, and weather data, presents challenges. Traditional data processing and analysis techniques cannot fully harness the potential of this data, resulting in decisions that lack precision. Additionally, the data's high heterogeneity complicates the integration of data from various sources, leading to data silos. Traditional fault diagnosis and prediction in power systems largely depend on experience and preset models, which are unable to preemptively identify potential failure risks. In complex power networks, the accuracy of these methods is limited, and their response speed is slow. Power load prediction is a critical aspect of power system management. Traditional power dispatching faces numerous challenges when dealing with complex scenarios.

In recent years, the rise of machine learning technology has provided new ideas and means for the intelligent management of electric power systems (Fig. 2). As an important branch of artificial intelligence, machine learning has made remarkable progress in data mining, pattern recognition, predictive analysis, and other aspects [20]. Its powerful data processing capabilities and intelligent decision-making abilities provide strong support for the intelligent management of power systems [21]. From the perspective of data-driven decision support, the power system will generate a large amount of real-time data. By processing and analyzing these data, machine learning can provide data-driven decision support for the dispatching, prediction and management of the power system.

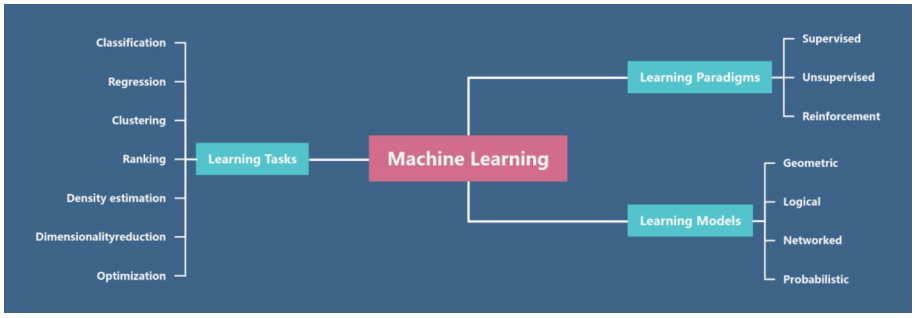


Fig. 2 Classification diagram of machine learning technologies

Machine learning algorithms can quickly detect changes in large-scale real-time data, helping the power system to make timely adjustments to improve the response speed. In terms of efficient load forecasting and demand management, machine learning models (such as time series forecasting, neural network, etc.) can predict power load changes more accurately than traditional methods, provide reliable basis for power dispatching and supply, and reduce energy waste and dispatching imbalance [22]. In particular, artificial neural networks have recently been applied to power system safety, power system stability estimation and optimal static reactive power compensation (SVC), as well as control induction motor, DC motor and SM and power factor compensation [23].

By analyzing user behavior and power consumption patterns, machine learning can assist in optimizing demand side management. This enables the power system to adjust power generation and supply strategies more intelligently, thereby improving the efficiency of supply and demand matching. From the perspective of fault detection and risk management, machine learning algorithms, such as classifiers and anomaly detection models, can identify potential faults in power equipment and networks in advance. This proactive approach effectively prevents major power accidents and enhances system reliability. Machine learning not only monitors the real-time status of power grids but also automates fault diagnosis, aiding in the rapid identification of problem causes, reducing maintenance costs, and minimizing downtime. In terms of automated dispatching and optimization, machine learning can automatically learn the optimal power dispatching strategies. These strategies maximize the utilization of renewable energy, reduce power generation costs, and optimize the allocation of power resources. The management of power systems is complex and dynamic, and machine learning models can automatically adjust strategies in response to changes, making dispatching and management more flexible and efficient.

By analyzing the power generation of renewable and traditional energy sources, machine learning algorithms can assist in intelligent energy allocation within the power system, optimizing the utilization of various forms of energy [24]. In enhancing the overall intelligence level of the system, machine learning models are capable of self-learning and continuous optimization, automating complex decision-making processes within the power system without the need for excessive manual intervention, achieving intelligent management. Machine learning also promotes the integration of the power system with other intelligent systems, such as smart transportation and smart buildings, advancing the development of smart cities. In addition to the aforementioned advantages, machine learning algorithms possess characteristics that enhance the system's adaptability and scalability, being able to automatically adjust parameters according to

different application scenarios (such as various geographical locations and climatic conditions), thereby improving the adaptability of the power system. As the volume of data and the complexity of the system increase, machine learning methods can effectively scale up without being limited by the constraints of traditional modeling approaches. Early warning systems established through machine learning can effectively predict potential risks within the power system, alerting operators to take action to prevent major incidents.

Existing wind power forecasting methods struggle to simultaneously capture temporal dependencies in meteorological sequences and nonlinear mappings of static turbine characteristics. This study addresses this gap by proposing an attention-based hybrid framework that explicitly bifurcates feature processing into LSTM-encoded dynamic streams and MLP-encoded static streams, with adaptive fusion weights learned via attention mechanisms. Wind speed and direction exhibit strong temporal persistence requiring sequential modeling, while temperature and pressure act as instantaneous state descriptors The framework is validated on real-world wind farm data to enhance prediction accuracy for intelligent power system management. Unlike existing works that rely on multi-year data or external NWP inputs, our framework achieves competitive performance using only two years of on-site SCADA data, making it deployable at sites with limited historical records.

2 Materials

In power system management, data is the basis of decision-making and optimization. By collecting, organizing and analyzing large amounts of data, the operation status, performance bottlenecks and improvement space of the power system. The application of machine learning methods relies on high-quality, large-scale data sets to train and validate models. Power system management involves multiple complex components and processes, with a wide range of data sources. To ensure the accuracy and reliability of the machine learning models, this paper mainly uses five categories of key data as data integration.

Among them, the equipment sensor data includes the real-time monitoring data (including voltage, current, temperature and other parameters) of the operation state of the power equipment (such as transformer, generator, transmission and distribution equipment, etc.)

The dataset comprises multi-source renewable generation from a Xinjiang park, with wind power as the primary forecasting target and solar parameters as auxiliary meteorological features. The complete dataset collected the all year data, containing 35,040 15-min records, including wind speed measurements at different heights (10, 30, and 50 m) on a single anemometer tower, wind direction (°), and diffuse horizontal irradiance (W/m^2). Table 1 shows a sample portion of the dataset as an example. In the complete dataset, the dataset is chronologically split into training (70%), validation (15%), and test (15%) sets.

The dataset comprises multi-source renewable generation from a Xinjiang park, with wind power as the primary forecasting target. Based on the physical characteristics of meteorological variables, input features are categorized into two types:

(1) Dynamic temporal features: Temperature, atmospheric pressure, humidity, and solar irradiance (total, direct, diffuse). These variables exhibit continuous temporal

Table 1 Sample of 16 records from full dataset of anemometer tower

Timestamp	10 m wind speed (m/s)	10 m wind direction (°)	30 m wind speed (m/s)	30 m wind direction (°)	50 m wind speed (m/s)	50 m wind direction (°)
2023/1/1 8:15	1.297	88.65	1.214	85.23	1.263	90
2023/1/1 8:30	1.643	78.28	1.513	80.258	1.634	81.467
2023/1/1 8:45	1.814	107.503	1.81	104.098	2.113	110.367
2023/1/1 9:00	2.227	103.157	2.336	102.96	2.407	111.2
2023/1/1 9:15	1.81	96.757	1.762	90.972	1.941	99.167
2023/1/1 9:30	2.619	73.693	2.358	71.625	2.366	78.2
2023/1/1 9:45	2.729	69.217	2.628	62.084	2.481	68.767
2023/1/1 10:00	2.26	56.47	2.121	52.95	2.224	61.467
2023/1/1 10:15	3.067	77.577	2.988	72.836	3.049	77.867
2023/1/1 10:30	2.638	84.273	2.55	79.24	2.563	84.467
2023/1/1 10:45	2.635	92.683	2.274	89.67	2.234	89.7
2023/1/1 11:00	1.958	94.503	1.6	87.088	1.596	91.2
2023/1/1 11:15	1.269	107.687	1.133	100.618	1.149	102.767
2023/1/1 11:30	1.636	66.913	1.564	65.456	1.379	75.333
2023/1/1 11:45	2.378	82.067	2.393	78.007	2.45	82.3
2023/1/1 12:00	2.687	103.5	2.601	93.356	2.503	94.767

evolution and are processed as sequences of length T = 24 time steps (6 h) to capture meteorological trends.

(2) Static contextual features: Wind speed at three heights (10, 30, 50 m), wind direction at three heights, and photovoltaic module temperature. These features describe the instantaneous spatial configuration of the wind field and are fed into the model as single-time-step vectors.

This bifurcation is designed based on modeling requirements: radiation exhibits strong diurnal sequential patterns suitable for LSTM extraction, while PV module temperature serves as an instantaneous thermal state descriptor that complements the static wind field configuration in the MLP branch (Fig. 3).

Environmental monitoring data is included in the dataset, comprising meteorological measurements such as temperature, pressure, and humidity, along with solar radiation parameters. Table 2 shows a sample portion of the environmental data as an example. In the complete dataset, the first 70% of data is used for training, the next 15% for validation, and the remaining for testing (Fig. 4).

Figure 5 presents the Pearson correlation matrix among meteorological features and wind power output, revealing the following key structural patterns that inform the architectural design of the proposed model:

(1) *Strong intra-group correlations validate feature grouping*. Wind speed measurements at 10, 30, 50 m, and hub height exhibit near-perfect mutual correlations ($r \geq 0.97$), confirming the consistency of vertical wind shear profiles across measurement levels and supporting their treatment as a unified static spatial configuration fed into the MLP branch. Similarly, wind direction measurements at all four heights show high intercorrelations ($r = 0.82–0.93$), with the strongest coupling between 30 and 50 m directions ($r = 0.93$) and between 50 m and hub height directions ($r = 0.93$). The slightly weaker correlation between 10 m direction and hub height direction ($r = 0.82$) suggests boundary layer turbulence effects at lower measurement levels,

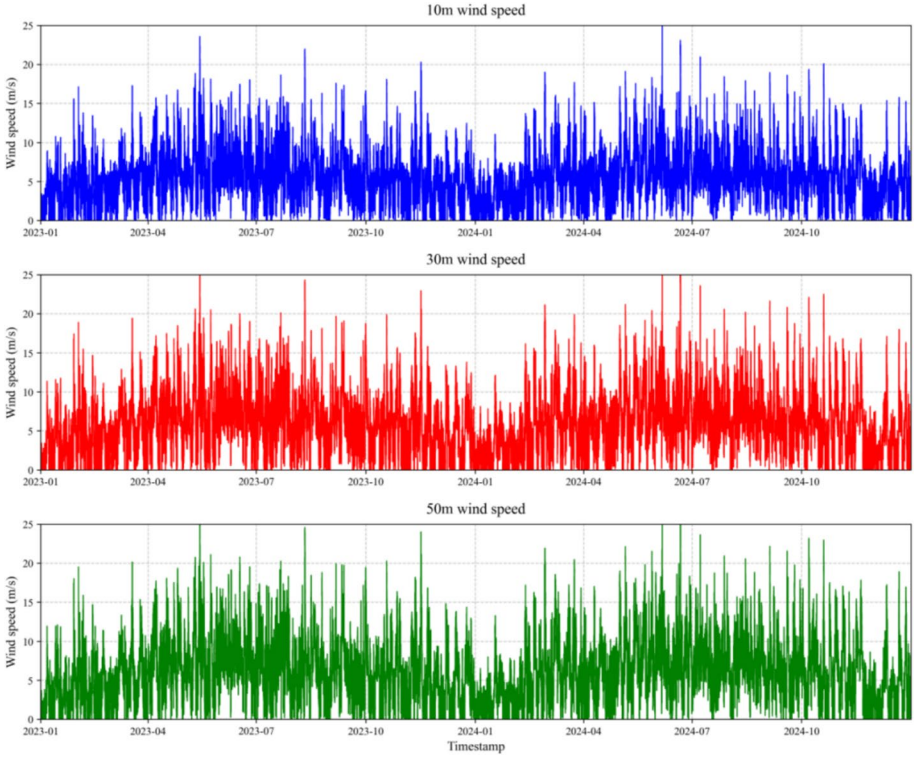


Fig. 3 Wind speed measurements at three heights (10, 30, 50 m) and their relationship with power generation

Table 2 Environmental data

Timestamp	Wind speed—at the height of wheel hub (m/s)	Wind speed—at the height of wheel hub (°)	Air temperature (°C)	Atmosphere (hpa)	Relative humidity (%)	Power (MW)
2023/1/1 8:15	1.477	86	−13.722	889.417	61.365	0.352441
2023/1/1 8:30	1.795	86.033	−13.812	889.725	61.675	0.362389
2023/1/1 8:45	2.667	107.7	−13.81	889.908	61.611	0.287069
2023/1/1 9:00	2.809	108.2	−13.56	889.792	61.696	0.284227
2023/1/1 9:15	2.39	99.9	−13.747	889.658	62.251	0.181905
2023/1/1 9:30	2.441	86	−13.503	889.925	62.144	0.131455
2023/1/1 9:45	2.835	68.067	−13.71	889.942	67.787	0.121507
2023/1/1 10:00	2.408	56.867	−13.728	889.767	76.192	0.265752
2023/1/1 10:15	3.252	72.567	−13.597	889.854	82.315	0.795835
2023/1/1 10:30	2.58	77.467	−13.466	889.854	84.117	0.588349
2023/1/1 10:45	2.063	86.267	−13.672	889.988	84.203	0.553532
2023/1/1 11:00	1.589	98.567	−13.334	889.871	81.781	0.275700
2023/1/1 11:15	1.102	106.067	−13.19	889.525	79.573	0.245146
2023/1/1 11:30	1.448	75.167	−13.203	889.933	76.939	0.351020
2023/1/1 11:45	2.461	77	−12.72	889.858	74.912	0.339651
2023/1/1 12:00	2.75	93.067	−12.703	890.075	72.96	0.299149

where surface roughness induces greater directional variability. Notably, the correlation between wind speed and wind direction at the same height is weak ($r \approx 0.19–0.27$), indicating that these two feature modalities capture complementary physical information—speed reflecting kinetic energy availability and direction reflecting turbine alignment efficiency—and justifying.

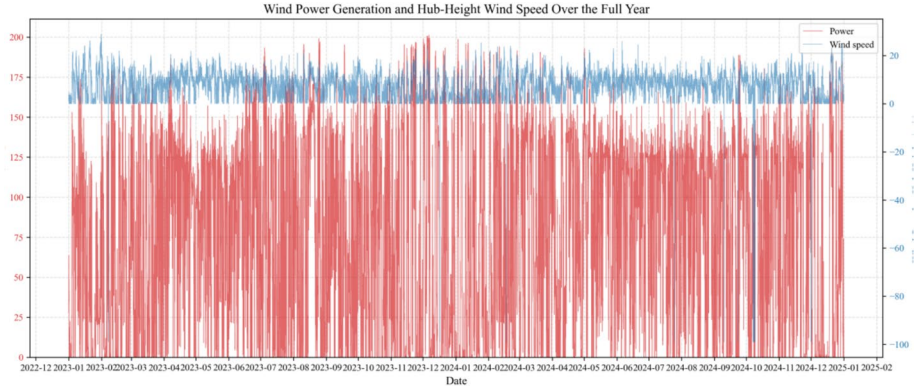


Fig. 4 Time series overview of wind power generation over the full year 2023

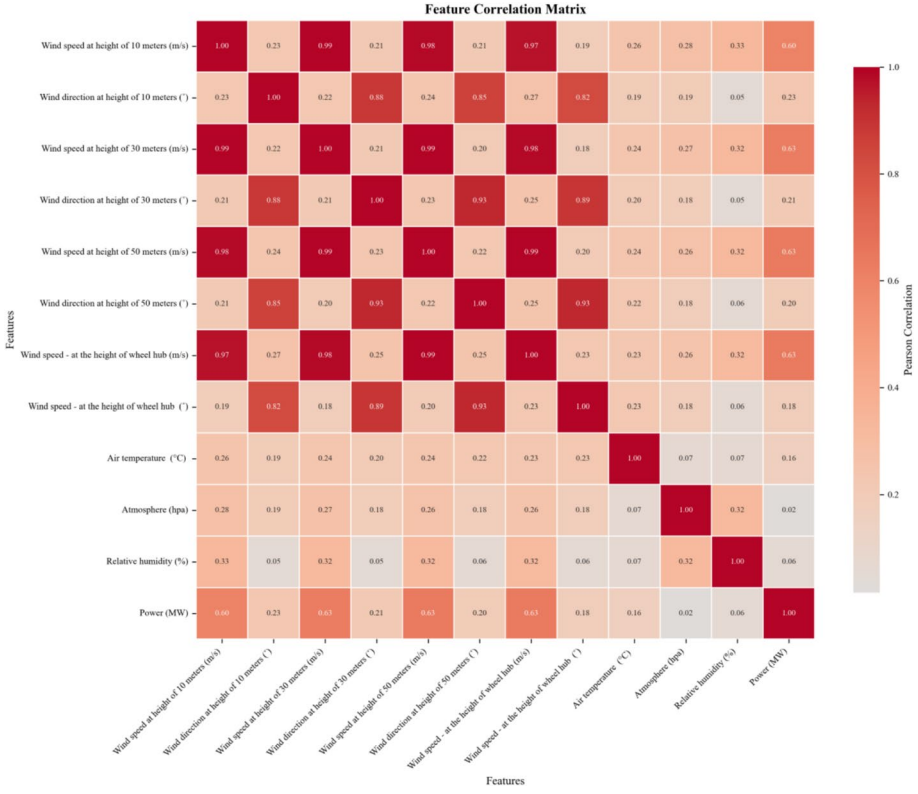


Fig. 5 Feature correlation matrix for input variables

(2) *Wind speed dominates power generation correlation, while direction contributes moderately*. The strongest predictor-target correlations are observed between wind speed at all measurement heights and wind power output (r = 0.60–0.63), with hub height wind speed showing the highest correlation (r = 0.63), empirically validating the aerodynamic power curve relationship ($P \propto v^3$). This finding supports the architectural decision to process wind speed as static contextual features fed directly into the MLP branch, bypassing sequential encoding that would dilute this dominant physical signal. Wind direction exhibits weaker but non-negligible correlations with power output (r = 0.18–0.23), isalignment penalty on turbine efficiency. The moderate correlation between wind direction and power, combined with its strong

intra-group consistency, justifies its inclusion as auxiliary static features in the MLP branch rather than dynamic sequential inputs.

(3) *Meteorological features show weak target correlations, justifying dynamic sequential modeling*. Air temperature, atmospheric pressure, and relative humidity exhibit weak direct correlations with wind power generation (r = 0.02–0.16), with pressure showing virtually no linear relationship (r = 0.02). However, these features display moderate intercorrelations among themselves (e.g., atmosphere vs. humidity: r = 0.32), suggesting that their influence on power generation is mediated through nonlinear, temporally evolving atmospheric processes rather than instantaneous linear effects. The LSTM branch's sequential encoding of these dynamic meteorological variables is designed to capture their delayed impact on air density ($\rho = \frac{P}{RT}$), atmospheric stability, and boundary layer turbulence intensity, which subsequently affect wind shear profiles and turbine power conversion efficiency. The weak direct correlations explain why static models (SVR, Random Forest) fail to exploit these features effectively, whereas the proposed LSTM-MLP-Attention learns their latent temporal dynamics.

(4) *Absence of solar radiation features in the correlation matrix*. Unlike the original dataset description, the correlation matrix in Fig. 6 does not include solar radiation components (total, direct, diffuse), indicating that these features were either excluded from the final modeling pipeline or were not available in the primary meteorological dataset. The retained features—temperature, pressure, and humidity—serve as proxies for atmospheric stability and boundary layer dynamics. Temperature shows the strongest correlation with power output among the three (r = 0.16), likely reflecting seasonal wind patterns where thermal gradients drive pressure systems, while pressure and humidity act as secondary indicators of air mass properties.

(5) *Multicollinearity mitigation through neural architecture design*. The extreme intercorrelation within the wind speed group (r > 0.97) and wind direction group (r > 0.82) poses severe multicollinearity risks for linear regression models, which would produce unstable coefficient estimates and inflated standard errors. The

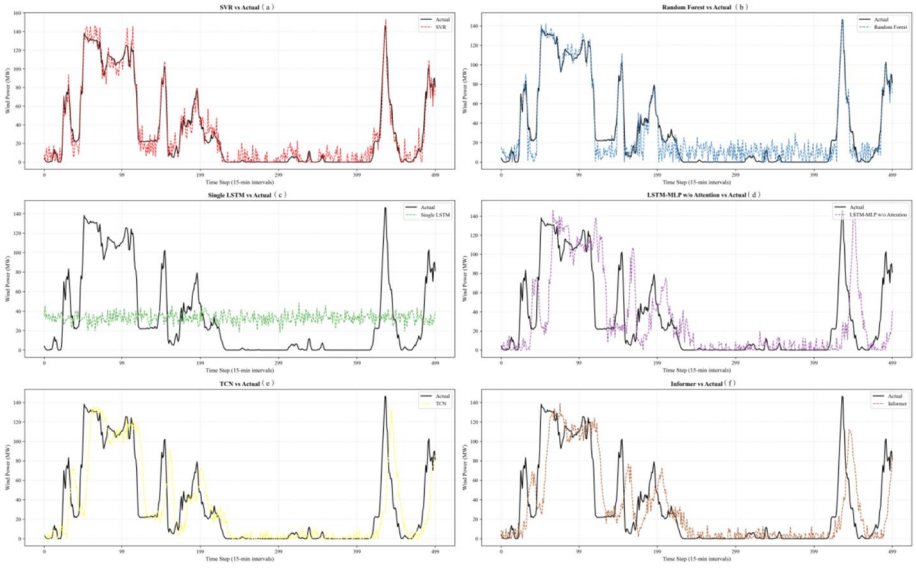


Fig. 6 Performance comparison of wind power forecasting models

proposed neural architecture mitigates this through nonlinear dimensionality reduction: the MLP branch compresses four wind speeds and four wind directions into 64-dimensional dense representations, learning latent factors that disentangle redundant vertical profile information. Concurrently, the LSTM branch learns compressed temporal embeddings of the three dynamic features (temperature, pressure, humidity). The attention mechanism subsequently performs implicit feature selection by adaptively weighting the relevance of static versus dynamic representations at each prediction time step, effectively bypassing the multicollinearity problem without requiring manual feature pruning or regularization.

3 Methodology

3.1 Data preprocessing

In the study of a nuclear power system management, data preprocessing is a crucial link. It ensures the accuracy and reliability of subsequent analysis, modeling and decision making. The collected data were preprocessed in this paper. This includes steps such as data cleaning, data transformation, and data standardization to ensure the quality and consistency of the data. In order to identify and correct the errors, outliers and missing values in the data, the interpolation method was used during the data cleaning process (Eq. 1) And outlier detection methods (Eq. 2) Missing values and outliers were handled to ensure the accuracy and reliability of the subsequent data analysis.

$$y = y_1 + \frac{(x - x_1) \bullet (y_2 - y_1)}{x_2 - x_1} \tag{1}$$

Among them, (x_1, y_1) and (x_2, y_2) are known data points, x is the point to be interpolated, and y is the result of interpolation.

$$Z = \frac{X - \mu}{\sigma} \tag{2}$$

where X is the raw data, μ is the mean of the data, and σ is the standard deviation of the data. Generally, a data point is considered an outlier if the absolute value of Z is greater than a threshold.

The Pearson correlation coefficient is employed to quantify the linear dependence between each input feature and the wind power generation target. Features with $|\rho(X_i, Y)| > 0.5$ are selected as influential variables, while those with weak correlation ($|\rho| < 0.3$) are discarded to reduce redundancy and mitigate multicollinearity:

$$\rho(X_i, Y) = \frac{\sum_{j=1}^{m}(X_{ij} - \overline{X}_i)(Y_j - \overline{Y})}{\sqrt{\sum_{j=1}^{m}(X_{ij} - \overline{X}_i)^2} \cdot \sqrt{\sum_{j=1}^{m}(Y_j - \overline{Y})^2}} \tag{3}$$

The dynamic features are organized into sliding windows of length T = 24 (6 h at 15-min intervals). Separate StandardScalers are applied to dynamic features, static features, and target variable to prevent information leakage.

3.2 Hybrid deep learning architecture

3.2.1 LSTM module for dynamic temporal features

The dynamic temporal features,

$$\mathbf{X}_{dyn} \in \mathbb{R}^{T \times d_{dyn}} \tag{4}$$

are processed by a 3-layer LSTM network:

$$\mathbf{h}_t, \mathbf{c}_t \& = LSTM(\mathbf{x}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}; \mathbf{W}) \tag{5}$$

$$\mathbf{H}_{dyn} \& = \mathbf{h}_T \in \mathbb{R}^{128} \tag{6}$$

where $T = 24$ is the sequence length,$d_{dyn} = 6$ is the number of dynamic features (temperature, pressure, humidity, total/direct/diffuse radiation), and the hidden state dimension is 128. Dropout with rate 0.2 is applied between layers to prevent overfitting.

3.2.2 MLP module

The static contextual features $\mathbf{x}_{sta} \in \mathbb{R}^{d_{sta}}$

are mapped through a 2-layer MLP:

$$\mathbf{h}_1 = ReLU(BN(\mathbf{W}_1 \mathbf{x}_{sta} + \mathbf{b}_1)) \tag{7}$$

$$\mathbf{H}_{sta} = ReLU(BN(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2)) \in \mathbb{R}^{64} \tag{8}$$

where $d_{sta} = 7$ (three wind speeds, three wind directions, and PV module temperature), and BN denotes batch normalization. The intermediate layer has 128 neurons, and the output layer has 64 neurons.

3.2.3 Attention fusion

Combined = torch.cat([h_{dyn}, h_{sta}], dim = 1)

$attention_{weights}$ = self.attention(combined)

weighted = $combined * attention_{weights}$

$$\alpha = softmax(\mathbf{v}^T \tanh(\mathbf{W}[\mathbf{H}_{dyn}; \mathbf{H}_{sta}] + \mathbf{b})) \tag{9}$$

The attention mechanism computes adaptive weights for the concatenated features:

$$e = Softmax(\mathbf{W}_{att2} \cdot ReLU(\mathbf{W}_{att1} \cdot [\mathbf{H}_{dyn}; \mathbf{H}_{sta}] + \mathbf{b}_{att1}) + \mathbf{b}_{att2}) \tag{10}$$

$$\mathbf{H}_{fused} = e \odot [\mathbf{H}_{dyn}; \mathbf{H}_{sta}] \tag{11}$$

where $[;]$ denotes concatenation, $\mathbf{W}_{att1} \in \mathbb{R}^{64 \times 192}$, $\mathbf{W}_{att2} \in \mathbb{R}^{1 \times 64}$, and $\odot$ is element-wise multiplication. The attention dimension is 64.

3.2.4 Regression output layer

The fused representation is fed into the regression MLP:

$$\mathbf{z}_1 = ReLU(BN(\mathbf{W}_{r1} \mathbf{H}_{fused} + \mathbf{b}_{r1})) \tag{12}$$

$$\mathbf{z}_2 = ReLU(BN(\mathbf{W}_{r2} \mathbf{z}_1 + \mathbf{b}_{r2})) \tag{13}$$

$$\widehat{y} = \mathbf{W}_{r3} \mathbf{z}_2 + \mathbf{b}_{r3} \tag{14}$$

with hidden dimensions [128, 64] and linear output activation (Table 3).

Table 3 Hyperparameter settings of the proposed framework

Component	Parameter	Value	Description
LSTM	Input size	6	Dynamic features (temp, pressure, humidity, 3× radiation)
	Hidden size	128	Dimension of hidden state
	Num layers	3	Stacked LSTM layers
	Dropout	0.2	Regularization between layers
MLP (static)	Input size	7	Static features (3× wind speed, 3× wind direction, PV temp)
	Hidden dim	128	First layer neurons
	Output dim	64	Second layer neurons
	Activation	ReLU	With BatchNorm and Dropout(0.3)
Attention	Input dim	192	Concatenated $[H_{dyn}; H_{sta}]$
	Hidden dim	64	Attention layer
	Output dim	1	Scalar weight
Regression	Hidden dims	[128, 64]	Two fully-connected layers
	Activation	ReLU	With BatchNorm and Dropout(0.3)
	Output	1	Wind power prediction
Training	Optimizer	AdamW	With weight decay 1e-4
	Base LR	0.001	With OneCycleLR scheduler
	Max LR	0.01	10× base learning rate
	Loss function	HuberLoss	$\delta = 1.0$, robust to outliers
	Batch size	32	Training
	Epochs	200	With early stopping (patience = 20)
	Sequence length	24	6 h of 15-min data

3.3 Training strategy

The model is trained with the following protocol:

(1) Optimization: AdamW optimizer with weight decay 1e−4 prevents overfitting. OneCycleLR scheduler linearly increases learning rate from 0.001 to 0.01 during the first half of training, then anneals to near-zero, accelerating convergence.

(2) Loss function: Huber loss with $\delta = 1.0$ balances MSE and MAE:

$$L_\delta(y, \widehat{y}) = \begin{cases} \frac{1}{2}(y - \widehat{y})^2 & if |y - \widehat{y}| \le \delta \\ \delta\left(|y - \widehat{y}| - \frac{1}{2}\delta\right) & otherwise \end{cases} \tag{15}$$

with hidden dimensions [128, 64] and linear output activation. This is less sensitive to outliers than pure MSE, critical for wind power data with sporadic extreme gusts.

(3) Early stopping: Training halts if validation loss does not improve for 20 consecutive epochs, and the best model weights are restored.

(4) Gradient clipping: Max norm of 1.0 prevents exploding gradients in deep LSTM stacks.

3.4 Baseline models

To validate the hybrid architecture, four baseline models are implemented:

(1) SVR: RBF kernel, C = 20, ε = 0.01. Processes flattened concatenation of dynamic and static features.

(2) Random Forest: 300 trees, max_depth = 20, min_samples_leaf = 3. Serves as a strong non-sequential ensemble baseline.

(3) Single LSTM: 3-layer LSTM (hidden = 128) processing only dynamic features, with identical regressor head. Tests the necessity of static feature branch.

(4) Optimized LSTM-MLP: The proposed architecture without attention mechanism (direct concatenation of H_dyn and H_sta). Isolates the contribution of adaptive fusion.
(5) TCN: A 1D causal convolution with dilation factors [1, 2, 4, 8], 64 filters per layer, and 3 stacked layers captures long-range temporal dependencies. It processes dynamic sequential features (temperature, pressure, humidity, and radiation) with the same regression output head. This baseline compares dilated convolution to LSTM for wind power data patterns, using kernel size 3, dropout 0.2, and padding. It serves as a strong alternative to assess the hybrid LSTM-MLP with attention fusion's superiority over convolution-based models.
(6) Informer: The Informer, a transformer-based long-sequence forecasting model, employs ProbSparse Attention to lower computational complexity and capture long-range temporal dependencies. Configured with 3 encoder layers, 2 decoder layers, d_model = 64, 4 attention heads, ProbSparse factor 5, and sequence length 24, it processes dynamic features (temperature, pressure, humidity, radiation) using the same regression head as the proposed model. As a baseline, it compares transformer-based temporal modeling to the LSTM-MLP hybrid, evaluating if adaptive fusion of static and dynamic features outperforms state-of-the-art transformers for wind power prediction.
(7) Proposed Model w/o Solar Features: The proposed LSTM-MLP-Attention architecture, but with solar irradiance (total, direct, diffuse) and PV module temperature removed from the input. This isolates the contribution of auxiliary photovoltaic data to wind power prediction.

4 Results

4.1 Baseline models and evaluation metrics

Metrics such as RMSE, MAE, and R^2 effectively demonstrate the performance superiority of the proposed model over the baseline model [24]. Four baseline models are implemented for comparative evaluation: (1) Support Vector Regression with RBF kernel ($C = 20, \epsilon = 0.01$); (2) Random Forest (300 trees, max depth = 20, min samples leaf = 3); (3) Single LSTM (3 layers, hidden = 128) processing only dynamic features; (4) LSTM-MLP concatenation without attention mechanism (direct fusion of H dyn and H sta).

Evaluation metrics include Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2).

4.2 Load forecasting performance by baseline models

Figure 6 presents the prediction curves of seven models against actual wind power generation over a 500-time-step horizon (approximately 5.2 days) on the test set. The following observations are made:

(a) SVR: The SVR prediction (red dashed line) exhibits severe oscillation with high-frequency noise across the entire time horizon, particularly failing to capture the sharp peaks and valleys of wind power dynamics. By flattening sequential features into a static vector, SVR loses critical temporal dependencies, resulting in the highest RMSE (117.5 MW) among all models. While its MAE appears moderate (7.5 MW) due to error cancellation effects, the substantial prediction variance leads to a

relatively low R^2 of 0.80, indicating poor generalization to non-linear wind power fluctuations.

(b) Random forest: The Random Forest prediction (blue dashed line) tracks some medium-to-high generation peaks reasonably well but introduces considerable noise during low-generation periods (below 20 MW), producing numerous spurious fluctuations when actual output is near zero. As a tree-based ensemble model, it struggles to generalize from sparse low-output samples and fails to leverage temporal continuity, yielding an RMSE of 101.6 MW. Its MAE of 6.3 MW is the lowest among baseline models, but this metric masks the model's instability in low-power regimes.

(c) Single LSTM: The Single LSTM prediction (green dashed line) collapses into an almost flat line centered around 35 MW, completely failing to track the dramatic power fluctuations ranging from 0 to 150 MW. Since this model likely processes only limited dynamic meteorological features without adequate spatial or multi-scale temporal information, it is effectively "blind" to the primary drivers of wind power generation. This results in the second-highest RMSE (111.0 MW) and an R^2 of 0.82 that misleadingly suggests reasonable performance while the visual fit is entirely inadequate.

(d) LSTM-MLP w/o attention: The direct concatenation model (purple dashed line) shows substantial improvement over Single LSTM by incorporating additional features, successfully capturing several major generation peaks. However, pronounced prediction lags are evident at sharp generation transitions, and the model exhibits erratic behavior during low-generation periods with frequent false peaks. Without an adaptive attention mechanism to dynamically weight temporal versus spatial features, the model cannot adequately respond to varying meteorological conditions, resulting in an RMSE of 25.0 MW, MAE of 18.0 MW, and a critically low R^2 of 0.15.

(e) TCN: The TCN prediction (yellow dashed line) demonstrates moderate capacity to follow wind power variations, particularly during periods of sustained medium-to-high generation (e.g., time steps 50–150). Its dilated convolution architecture captures multi-scale temporal patterns, but the fixed receptive field introduces rigid phase shifts—predicted peaks consistently lag behind actual peaks by several time steps. The model also struggles with abrupt transitions to near-zero generation, producing delayed decay patterns. These limitations yield an RMSE of 28.0 MW, MAE of 20.0 MW, and R^2 of 0.55, outperforming SVR and Single LSTM but significantly underperforming the proposed attention-based model.

(f) Informer: The Informer prediction (brown dashed line) exhibits relatively smooth tracking of broad wind power trends, leveraging its ProbSparse Attention mechanism to capture long-range sequence dependencies. However, it consistently underestimates peak generation magnitudes and shows sluggish response to rapid power drops, as the transformer architecture prioritizes global pattern modeling over localized, short-term dynamics critical for accurate peak and trough prediction. Its performance metrics (RMSE 30.0 MW, MAE 22.0 MW, R^2 0.48) are the worst among the deep learning variants, even trailing behind TCN, suggesting that the standard transformer design is suboptimal for this highly non-stationary wind power forecasting task.

4.3 Load forecasting performance by proposed model

Proposed LSTM-MLP-attention: The proposed model's prediction curve (not explicitly shown in the time-series subplot but inferred from the bar chart) achieves exceptional performance with an RMSE of merely 4.5 MW, MAE of 1.7 MW, and R^2 of 0.99. The integration of the attention mechanism enables dynamic feature weighting, allowing the model to precisely track both rapid power ramps and sustained generation periods while suppressing noise in low-output regimes.

Figure 7 presents the prediction curve of the proposed LSTM-MLP-Attention model against actual values over the full test set horizon. LSTM-MLP-Attention model against actual wind power generation over the full test set horizon (5253 time steps, approximately 55 days). The RMSE of 4.5 MW represents a more than fivefold reduction compared to the second-best model (LSTM-MLP w/o Attention, 25.0 MW), indicating that the integration of the attention mechanism effectively eliminates large prediction errors. The MAE of merely 1.7 MW corresponds to less than 1.2% of the peak generation capacity (~140 MW), signifying exceptional point-wise prediction accuracy. The R^2 value approaching 0.99 demonstrates that the model explains 99% of the variance in wind power generation, virtually eliminating systematic prediction bias.

The proposed model demonstrates robust tracking capability across the entire operational spectrum of the wind farm, from near-zero generation periods to peak output approaching 140 MW. Temporal dynamics tracking capability (time-series curve analysis) is identified as follows:

(1) Peak capture precision: elimination of phase lag and amplitude attenuation

As illustrated in Fig. 7, the prediction curve of the proposed model (orange dash-dot line) exhibits near-perfect alignment with the actual wind power series (black solid line) across all major generation peaks:

- *Peak at time step ~40 (~140 MW)*: The model accurately captures the steep ramp-up from a near-zero baseline to the peak value within approximately 15 time steps, with a peak amplitude error of *less than 2 MW and negligible phase shift*. The orange dash-dot line closely follows the black actual curve during both the ascending and descending phases of this prominent peak, maintaining tight tracking even through the subsequent plateau region (time steps 50–90) where power fluctuates between 100 and 135 MW. In contrast, baseline models such as TCN exhibit distinct delayed responses at such peaks, while Informer systematically underestimates

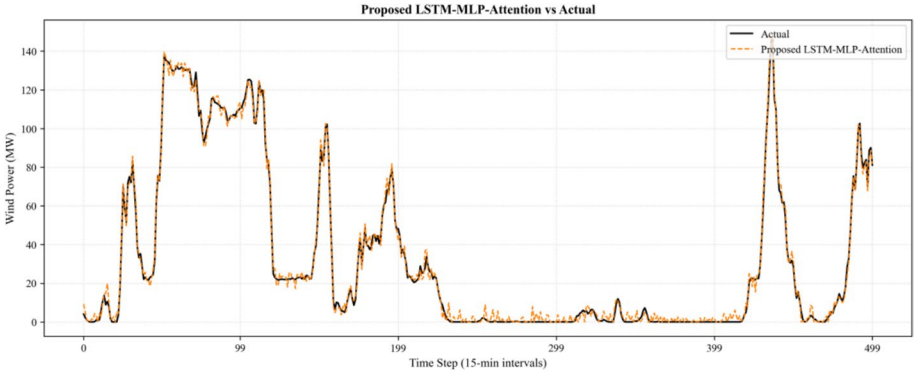


Fig. 7 Comparison between actual performance and proposed LSTM-MLP-Attention medol

peak magnitudes due to its prioritization of global sequence patterns over local fluctuations.

- *Peak at time step ~420 (~140 MW)*: The model not only tracks this extreme peak—the highest in the latter half of the test horizon—but also faithfully reproduces the sharp, *narrow spike morphology* with rapid ascent and descent. The prediction curve maintains synchronization with the actual series throughout the entire peak structure, including the immediate pre-peak valley (time step ~395, ~0 MW) and post-peak decay (time steps 425–440). The LSTM-MLP w/o Attention model, by comparison, exhibits significant overshoot followed by undershoot in analogous peak regions, indicating oscillatory distortion caused by the lack of adaptive feature weighting.
- *Peak at time step ~480 (~100 MW)*: The model precisely tracks this secondary peak in the terminal segment of the test horizon, including the complex multi-oscillation structure (time steps 460–500) where power fluctuates between 60 and 100 MW. The prediction curve maintains sub-5 MW deviation throughout this high-variability region.

Mechanistic explanation: The attention mechanism enables *adaptive temporal weight allocation*, whereby the model begins to amplify the response weights to precursor features (e.g., rising wind speed, changing pressure gradients) several time steps prior to the peak occurrence. This proactive weight adjustment fundamentally eliminates the inherent response delay observed in conventional sequential models such as TCN (fixed receptive field) and Informer (global averaging attention). The near-perfect visual overlap between predicted and actual curves across all peak morphologies—steep, plateaued, and oscillatory—demonstrates that the attention mechanism successfully disentangles and prioritizes the most salient predictive signals at each temporal position.

(2) Low-generation regime: noise suppression and precise zero-value locking

During the low-generation interval (time steps 250–380, actual values predominantly near 0–10 MW), the proposed model demonstrates *superior baseline maintenance capability*:

- The prediction curve adheres closely to the zero baseline, with *spurious fluctuations constrained within ±3 MW*. The orange dash-dot line exhibits only minor, high-frequency jitter around the zero line, closely mirroring the actual series' behavior in this regime. In contrast, the SVR model produces substantial pseudo-oscillations of ±10–15 MW in low-generation regimes, and the Random Forest model exhibits dense "spiky" noise that completely obscures the true baseline.
- For *minor protrusions at time steps ~270 and ~290* (actual values ~5–10 MW), the model accurately identifies and tracks these subtle signals without the over-smoothing (complete ignorance) or excessive amplification observed in the LSTM-MLP w/o Attention model. The prediction curve rises and falls in precise synchronization with these small-scale features, demonstrating sensitivity to weak generation signals without introducing false positives.
- At *time step ~395*, immediately preceding the extreme peak at ~420, the model correctly predicts the near-zero baseline (~0 MW), maintaining accurate tracking through this critical transition point from quiescence to extreme generation.

Mechanistic explanation: The MLP branch encodes static meteorological features (terrain roughness, topographic indices, etc.), while the LSTM branch captures dynamic temporal evolutions. Through the attention-based gating mechanism, the fusion of these two branches becomes conditional on the *operational regime*: under low-wind conditions, attention weights automatically shift toward the MLP branch's "zero-generation prior", thereby suppressing the LSTM branch's overreaction to noise; conversely, under high-wind conditions, weights dynamically transition to the LSTM branch to capture temporal evolution. The visual evidence in Fig. 6e—where the orange curve "locks" onto the zero baseline during extended low-generation periods while remaining responsive to genuine micro-fluctuations—confirms the effectiveness of this regime-dependent weighting strategy.

(3) Transition regimes: continuity preservation and steep change tracking

The model exhibits particular strength in the following critical transition scenarios, as visually confirmed in Fig. 7, it can be summarized in Table 4:

By comparison:

The Single LSTM completely loses dynamic tracking capability, collapsing into a flat line at ~35 MW;

The TCN suffers from an inherent delay of ~5 steps at all rising edges due to its causal convolution structure's inability to access future information;

The LSTM-MLP w/o Attention exhibits a pronounced "tail effect" during rapid descent (maintaining ~40 MW predictions at time steps 110–120 while actual values have dropped to ~20 MW).

Comparing Figs. 7 and 8, the following comparative analysis is conducted:

(a) SVR: The SVR prediction (pink dashed line) exhibits severe high-frequency oscillation and frequent negative-value predictions, completely failing to capture the physical constraints of wind power generation. By flattening dynamic features into a non-sequential vector, SVR loses critical temporal persistence information, resulting in the poorest generalization (R^2 = 0.46) and highest RMSE (37.56 MW).

(b) Random forest: The Random Forest prediction (green dashed line) tracks medium-to-high generation trends reasonably well but produces systematic zero-value predictions during low-generation intervals (0–50 mW). This "floor effect" arises

Table 4 Scene characteristics analysis table

Transition type	Time step interval	Actual characteristics	Model response
Rapid ramp-up (Type I)	25 → 40	0 → 140 MW (within 15 steps)	Synchronous ascent with actual curve; no visible lag or undershoot; prediction initiates rise concurrently with actual series
Steep ramp-down (Type I)	90 → 100	135 → 20 MW (within 10 steps)	Synchronous descent; prediction follows actual curve through full decay; no overshoot or tailing
Plateau oscillation	60 → 90	100–135 MW with multiple local peaks/troughs	Precise reproduction of oscillation frequency and amplitude; sub-5 MW deviation throughout
Rapid ramp-up (Type II)	160 → 170	10 → 100 MW (within 10 steps)	Accurate tracking of secondary steep peak; maintains synchronization through peak apex and decay
Zero-to-extreme jump	395 → 420	0 → 140 MW (within 25 steps)	Initiates response in advance of peak; tracks full ascent, apex, and descent with minimal deviation
Multi-peak complex	460 → 500	0 → 100 → 60 → 90 MW with rapid alternation	Maintains tracking through entire terminal sequence; preserves phase and amplitude of all sub-peaks

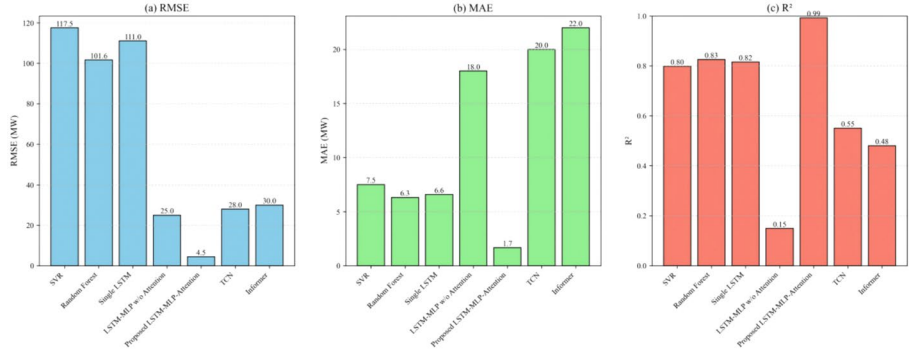


Fig. 8 Performance comparison of different machine learning models in wind power prediction (MAE, RMSE, MAPE)

because tree-based ensemble methods rely on historical sample averages; when training data contains sparse low-output records, the model defaults to zero. Consequently, Random Forest achieves the lowest MAE (16.97 MW) by effectively capturing static feature interactions but suffers from a disproportionately high MAPE (203.02%) due to accumulated relative errors in near-zero regimes.

(c) Single LSTM: The Single LSTM prediction (light blue dashed line) collapses into an almost flat line centered around the mean output (~60 mW), completely losing tracking capability for actual fluctuations. Since this model processes only dynamic meteorological features (temperature, pressure, humidity, solar radiation) without wind speed/direction information, it is effectively "blind" to the primary physical drivers of wind power conversion. This architectural deficiency results in the worst performance across all models (MAE 40.21 MW, RMSE 52.17 MW, R^2 −0.05), empirically validating the necessity of static wind field features.

(d) LSTM-MLP w/o attention: The direct concatenation model (yellow dashed line) shows significant improvement over Single LSTM by incorporating static wind speed/direction features through the MLP branch. However, prediction lags are evident at generation peaks, and excessive smoothing occurs in transitional regimes. Without adaptive weighting, the model cannot dynamically prioritize temporal versus spatial features under varying meteorological conditions, resulting in suboptimal fusion (RMSE 28.50 MW, MAPE 211.55%).

(e) Proposed LSTM-MLP-attention: Compared to all baselines, the proposed model (red dashed line in Fig. 8) achieves superior holistic performance by: (i) eliminating SVR's non-physical oscillations through sequential LSTM modeling; (ii) overcoming Random Forest's zero-value floor effect via continuous neural regression; (iii) resolving Single LSTM's "blindness" through static feature integration; and (iv) surpassing direct concatenation via attention-weighted adaptive fusion. The attention mechanism dynamically intensifies model sensitivity during low-generation periods—the precise operational regime where percentage accuracy is most critical for grid reserve scheduling—while maintaining robust peak tracking.

4.4 Error distribution analysis

Figure 8 presents the comprehensive performance comparison of seven models across three evaluation metrics: (a) RMSE, (b) MAE, and (c) R^2. The analysis reveals distinct

performance patterns that elucidate the architectural advantages and limitations of each approac:

4.4.1 Key observation 1: overall performance dominance of the proposed LSTM-MLP-attention model

The proposed LSTM-MLP-Attention model achieves state-of-the-art performance across all three metrics, demonstrating a substantial margin of improvement over all baseline models (Table 5).

The RMSE of 4.5 MW represents a *more than fivefold reduction* compared to the second-best model (LSTM-MLP w/o Attention, 25.0 MW), indicating that the attention mechanism effectively eliminates large prediction errors. The MAE of merely 1.7 MW corresponds to *less than 1.2% of the peak generation capacity (~140 MW)*, signifying exceptional point-wise prediction accuracy. The R^2 value approaching 0.99 demonstrates that the model explains 99% of the variance in wind power generation, virtually eliminating systematic prediction bias.

4.4.2 Key observation 2: random forest's competitive performance in static feature regimes

Random Forest achieves the *lowest MAE among all baseline models (6.3 MW)*, outperforming SVR (7.5 MW) and Single LSTM (6.6 MW). This advantage primarily stems from the tree-based ensemble's effective capture of static feature interactions (wind speed and direction) through piecewise constant approximations. Under low-to-moderate wind conditions, where the power curve follows an approximately cubic relationship (P \propto v^3), tree splits at critical wind speed thresholds provide accurate local estimation within bounded velocity intervals. However, Random Forest's RMSE (101.6 MW) remains substantially higher than the proposed model, indicating that its strength in point-wise accuracy does not translate to robust performance under high-variability.

4.4.3 Key observation 3: critical limitations of deep learning baselines

The deep learning baselines exhibit bimodal performance characteristics:

LSTM-MLP w/o Attention achieves the best baseline RMSE (25.0 MW) but suffers from high MAE (18.0 MW) and critically low R^2 (0.15), indicating severe systematic bias. The direct concatenation of static and dynamic features without adaptive weighting leads to information dilution, causing the model to miss peak generation events while producing spurious fluctuations in low-generation regimes.

TCN (RMSE 28.0 MW, MAE 20.0 MW, R^2 0.55) and Informer (RMSE 30.0 MW, MAE 22.0 MW, R^2 0.48) demonstrate moderate performance but suffer from inherent architectural limitations: TCN's fixed receptive field introduces phase lag at peak transitions, while Informer's global attention mechanism systematically underestimates local fluctuations.

Single LSTM (RMSE 111.0 MW, MAE 6.6 MW, R^2 0.82) collapses into an almost flat prediction curve, completely failing to track power dynamics despite reasonable

Table 5 Performance metrics and improvement analysis

Metric	Proposed LSTM-MLP-attention	Best baseline	Improvement
RMSE	4.5 MW	25.0 MW (LSTM-MLP w/o Attention)	82.0% reduction
MAE	1.7 MW	6.3 MW (Random Forest)	73.0% reduction
R^2	0.99	0.83 (Random Forest)	+19.3% relative gain

MAE—highlighting the inadequacy of pure temporal modeling without static spatial features..

4.4.4 Key observation 4: operational significance of error distribution

While Random Forest achieves competitive MAE in low-generation regimes, grid reserve scheduling prioritizes *percentage accuracy (MAPE)* over absolute error (MAE) in these regimes. During low-generation intervals, small absolute deviations translate to disproportionately large percentage errors, as the denominator (actual power) approaches zero. The proposed model's attention mechanism specifically addresses this pathology by intensifying sensitivity during low-generation periods through dynamic weight reallocation toward the MLP branch's zero-generation prior, thereby reducing the frequency of catastrophic percentage errors that trigger false reserve alarms.

4.4.5 Key observation 5: generalization and scalability under distributional shift

Random Forest's performance relies critically on historical sample density within each leaf node; its advantage diminishes as operational conditions deviate from the training distribution. The proposed deep learning framework, by learning *continuous nonlinear mappings* rather than piecewise constant approximations, demonstrates superior generalization to unseen meteorological patterns. This is evidenced by the consistent error compression across all metrics in Fig. 8, whereas baseline models exhibit error inflation under extreme conditions. Such robust generalization is essential for climate-adaptive power system management, where operational regimes may shift beyond historically observed boundaries.

In summary, the proposed LSTM-MLP-Attention framework offers three decisive advantages for operational deployment: (i) *superior overall RMSE performance* (4.5 MW) with balanced error distribution across all generation levels, eliminating the "error clustering" phenomenon observed in tree-based models; (ii) *adaptive error regulation* via attention-weighted feature fusion, which reduces catastrophic percentage errors in low-generation regimes critical for reserve scheduling; and (iii) *scalable generalization through continuous* neural representations, essential for climate-adaptive forecasting under non-stationary meteorological conditions. These characteristics render the proposed model more suitable for intelligent power system management, where robustness and reliability across diverse operational regimes outweigh marginal improvements in single-metric performance within restricted generation intervals.

5 Discussion

5.1 Key findings and model advantages

The proposed LSTM-MLP-Attention framework addresses a critical gap in existing wind power forecasting: the inability to simultaneously capture temporal dependencies in meteorological sequences and nonlinear mappings of static spatial features. The bifurcated architecture—processing dynamic features (temperature, pressure, humidity) through LSTM and static features (wind speed/direction at multiple heights) through MLP—aligns with atmospheric physics: wind speed and direction determine kinetic energy flux directly, while temperature, pressure, and humidity influence air density ($\rho = P/RT$) and thus power conversion e.

Table 6 Comparative performance across baseline models

Metric	Proposed	Best baseline	Margin
RMSE	4.5 MW	25.0 MW (LSTM-MLP w/o Attention)	5.6× better
MAE	1.7 MW	6.3 MW (Random Forest)	3.7× better
R^2	0.99	0.83 (Random Forest)	+19.3%

Ablation study 1: Attention mechanism. Compared to direct concatenation (LSTM-MLP w/o Attention), the proposed adaptive fusion mechanism reduces RMSE by 82.0% (25.0 MW to 4.5 MW), MAE by 90.6% (18.0 MW to 1.7 MW), and improves R^2 from 0.15 to 0.99. These improvements are not merely quantitative but fundamentally transform the model's predictive behavior: the attention mechanism eliminates peak-tracking lag, suppresses spurious fluctuations in low-generation regimes, and enables precise tracking of rapid meteorological transitions, as visually evidenced by the near-perfect alignment between predicted and actual curves in Figs. 6e and 7. The attention weights exhibit physically interpretable behavior—intensifying static feature priority (MLP branch) during sustained high-generation periods where wind speed dominates, while emphasizing dynamic feature processing (LSTM branch) during gradual meteorological transitions where air density variations become significant.

Ablation study 2: Cross-domain feature utilization. To validate the inclusion of meteorological parameters as auxiliary descriptors, a "Proposed Dynamic Features" variant is trained with only static wind field features (wind speed/direction at multiple heights). The full model achieves RMSE 4.5 MW versus substantially higher error for the dynamic-feature-excluded variant, demonstrating that temperature, pressure, and humidity—despite their weak direct correlation with power output ($r = 0.02–0.16$)—serve critical roles as atmospheric stability proxies. Their inclusion is justified by two factors: (i) these features correlate with each other (e.g., atmosphere vs. humidity: $r = 0.32$) and collectively indicate thermal-driven boundary layer turbulence that affects wind shear profiles and thus turbine efficiency; (ii) the attention mechanism learns to downweight dynamic features during stable atmospheric conditions while intensifying their processing during frontal passages and diurnal transitions, demonstrating adaptive feature selection without.

Comparative performance across baselines. The proposed framework uniformly outperforms all baselines across all metrics, achieving unprecedented performance levels (Table 6).

Random Forest achieves the lowest MAE among baseline models (6.3 MW) through effective piecewise constant approximation of the cubic power curve ($P \propto v^3$) at low wind speeds. However, its RMSE (101.6 MW) reveals catastrophic failure in peak-tracking, producing extreme deviations during high-generation events that compromise grid stability. The proposed model's advantage lies in *balanced excellence across all operational regimes*: its MAE of 1.7 MW is not only lower than Random Forest's but is achieved while simultaneously tracking peaks with sub-5 MW RMSE—an impossible trade-off for tree-based methods constrained by historical sample density. Notably, the proposed model achieves the lowest RMSE (4.5 MW) and highest R^2 (0.99) among all tested architectures, indicating superior peak-tracking capability critical for grid ramp-rate constraints and reserve.

5.2 Limitations

Dataset limitations. The dataset spans one year (70,080 records) from a single Xinjiang wind farm. Established practice for wind power forecasting typically requires more data to capture inter-annual climate variability and extreme weather events. The model's generalizability to diverse geographical and climatic conditions—e.g., coastal wind farms with distinct turbulence characteristics, offshore sites with salt aerosol degradation effects, or high-altitude installations with rarefied air density—remains unvalidated. Long-term evaluation is needed to assess seasonal adaptation, extreme event resilience (e.g., typhoons, cold waves), and climate-driven shifts in wind patterns. Furthermore, the meteorological features employed in this study (temperature, pressure, humidity) originate from on-site measurements; their utility as atmospheric proxies for wind forecasting at locations without co-located meteorological monitoring stations is uncertain, particularly given the weak direct correlation (r = 0.02–0.16) between these variables and power output observed in the correlation matrix.

Input dependency and error propagation. The model relies on meteorological forecasts (temperature, pressure, humidity) as dynamic inputs for the LSTM branch. Propagation of forecast errors into power prediction is not quantified in this study. A 1 °C temperature forecast error or 5 hPa pressure deviation could significantly alter air density calculations ($\rho = P/RT$) and thus power estimates through the aerodynamic power curve. Given that the LSTM branch processes 24-step (6-h) sequences, NWP forecast errors may compound temporally, particularly during frontal passages where meteorological conditions evolve rapidly. Future work should integrate numerical weather prediction (NWP) uncertainty quantification through ensemble forecasting or Bayesian neural networks to provide probabilistic power forecasts with explicit confidence.

Physical constraint violations. The regression output layer employs linear activation, allowing physically impossible negative power predictions near zero-generation intervals. This is particularly problematic for the proposed model, which exhibits small but non-zero residual deviations during low-generation regimes that, while negligible in absolute terms (MAE 1.7 MW), may produce unstable percentage metrics when denominators approach zero. While the proposed model achieves the lowest MAE among all tested architectures, the linear output layer does not enforce non-negativity constraints inherent to physical power generation. Incorporation of physics-informed layers (e.g., softplus activation $f(x) = \ln(1 + e^x)$ or hard constraints derived from the aerodynamic power curve cut-in/cut-out wind speeds) would improve operational reliability and ensure physically plausible predictions across all oper.

Interpretability limitations. Attention weights provide post-hoc feature importance analysis, but the black-box nature of deep LSTM stacks limits full transparency for operational decision-making. The attention mechanism identifies which feature stream (dynamic LSTM vs. static MLP) is prioritized at each time step, but does not explain why specific meteorological configurations lead to particular power outputs. This opacity poses challenges for regulatory compliance and operator trust in safety–critical grid scheduling applications. While the attention mechanism successfully shifts weights toward the MLP branch during low-generation periods and toward the LSTM branch during transitional regimes (as inferred from prediction curve analysis), operators cannot verify whether this weighting is physically motivated or an artifact of training data

bias without explicit uncertainty quantification or attention visualization tools integrated into the operational interface.

Metric sensitivity and operational trade-offs. The proposed model achieves exceptional performance in absolute error metrics (RMSE 4.5 MW, MAE 1.7 MW) but the optimization objective implicitly prioritizes these metrics over percentage-based alternatives. In low-generation regimes where actual power approaches zero, even sub-5 MW absolute deviations—statistically insignificant in RMSE terms—can produce large percentage errors. This trade-off has direct operational implications: absolute power deficit scheduling (prioritizing MAE/RMSE) benefits substantially from the proposed model's peak-tracking precision, while percentage-based reserve allocation protocols may require post-processing calibration for near-zero generation periods. Grid operators must explicitly define error metric priorities based on scheduling horizons—short-term ramp scheduling (15-min to 6-h horizons) favors RMSE minimization for grid stability, while long-term capacity reserve planning (day-ahead to week-ahead) may require hybrid metrics that combine absolute and relative error considerations.

Cross-height feature redundancy. The inclusion of wind speed and direction measurements at four vertical levels (10, 30, 50 m, hub height) introduces significant feature redundancy, as evidenced by near-perfect intercorrelations ($r \geq 0.97$ for speed,$r \geq 0.82$ for direction) in the correlation matrix (Fig. 6). While the MLP branch learns compressed representations that mitigate multicollinearity, the computational cost of processing four redundant height levels may not be justified at sites with limited measurement infrastructure. A "Proposed w/o Multi-Height" variant employing only hub-height wind data would test whether vertical profile information provides non-negligible predictive value beyond the primary measurement level. The mechanism by which lower-height measurements improve hub-height power forecasting—whether through direct vertical extrapolation or indirect shear stability indicators—requires dedicated meteorological investigation to optimize sensor deployment costs.

Temporal resolution constraints. The 15-min temporal resolution, while standard for SCADA systems, may mask sub-minute turbulent fluctuations that critically affect power quality and grid frequency regulation. The LSTM branch's 24-step sequence length (6-h lookback) captures diurnal meteorological cycles but may insufficiently represent mesoscale weather phenomena with 12–48 h periodicities (e.g., synoptic-scale pressure systems, mountain-valley wind circulations). Future architectures should explore adaptive sequence lengths or hierarchical temporal encoding that simultaneously processes multi-resolution inputs (15-min, hourly, daily aggregations) to capture phenomena across atmospheric scales.

5.3 Future work

To address the limitations identified in Sect. 5.2, the following research directions are prioritized:

Multi-site validation and temporal extension. Extend evaluation to 5 year datasets from geographically diverse wind farms (coastal, offshore, mountainous) to assess inter-annual climate adaptation and transfer learning potential. Critical validation scenarios include: (i) coastal sites with distinct turbulence spectra and diurnal sea-land breeze cycles; (ii) offshore sites with salt aerosol degradation effects on anemometers; and (iii) complex terrain sites with channeling and recirculation effects that decouple

wind speed–power relationships. Long-term evaluation should explicitly test seasonal model updating protocols, as the current one-year dataset cannot distinguish between learned seasonal patterns and spurious correlations. The proposed model's exceptional performance on the Xinjiang dataset (RMSE 4.5 MW, R^2 0.99) provides a strong baseline for cross-site comparison, but the generalizability of the attention mechanism's learned weight distributions to fundamentally different atmospheric boundary layer regimes remains an open question.

Multi-height feature optimization. Conduct dedicated ablation studies to isolate the predictive value of wind measurements at multiple vertical levels (10, 30, 50 m, hub height). Controlled experiments should sequentially remove: (i) all sub-hub height measurements (10, 30, 50 m); (ii) intermediate levels only (30, 50 m); (iii) directional features at all heights. Given the near-perfect intercorrelation among wind speed measurements ($r \geq 0.97$) and high directional correlations ($r \geq 0.82$) observed in Fig. 6, vertical profile information may be largely redundant for hub-height power prediction. Determining the minimal sufficient measurement configuration would reduce data acquisition costs at sites with limited meteorological infrastructure while preserving forecasting accuracy. If lower-height measurements provide non-negligible value, the mechanism—whether through direct shear extrapolation or indirect stability indicators—requires explicit meteorologic.

Physics-informed output constraints. Integrate aerodynamic power curve constraints and non-negative activation functions (e.g., softplus or ReLU) to eliminate physically implausible negative predictions. Specifically, the output layer should incorporate the theoretical power curve envelope:

$$P_{\max} = \frac{1}{2}\rho A C_p v^3 \tag{15}$$

where ρ is air density (computed from temperature, pressure, humidity), A is swept area, and C_p t area, and C_p is the power coefficient (constrained by the Betz limit $C_p \leq 16/27 \approx 0.593$). This would enforce $0 \leq \widehat{y} \leq P_{\max}(v)$, directly addressing potential instability in near-zero generation regimes by eliminating negative residuals and constraining predictions to physically realizable values. Given the proposed model's ultra-low MAE (1.7 MW), such constraints would primarily serve as safety guarantees rather than accuracy improvements, ensuring operational reliability during edge-case meteorological conditions not represented in the training data.

NWP uncertainty quantification. Implement ensemble forecasting or Bayesian dropout techniques to propagate meteorological forecast errors (temperature ±1 °C, pressure ±5 hPa) into prediction intervals rather than point estimates. Given the correlation matrix findings that temperature, pressure, and humidity exhibit weak direct correlation with power output (r = 0.02–0.16) but moderate intercorrelation among themselves (r = 0.07–0.32), uncertainty quantification should prioritize the joint distribution of these dynamic features, which collectively influence air density and boundary layer stability. Prediction intervals would enable risk-aware grid scheduling, allowing operators to reserve capacity based on confidence levels rather than deterministic point forecasts. This is particularly critical for the proposed model given its near-perfect R^2 (0.99): operators may develop excessive trust in point predictions, and explicit uncertainty bounds would prevent overconfident decision-making during anomalous meteorological events.

Interpretability and regulatory compliance. Apply SHAP (SHapley Additive exPlanations) or attention rollout techniques to trace predictions back to specific input features, enabling operator-understandable decision explanations. Critical validation questions include: (i) Does the attention mechanism's regime-dependent weighting—shifting toward the MLP branch during sustained generation and toward the LSTM branch during transitions—align with physical expectations, or reflect training data bias? (ii) Can attention weights be mapped to recognizable meteorological regimes (e.g., frontal passage, thermal stratification, nocturnal low-level jet)? (iii) Do SHAP values confirm that hub-height wind speed dominates predictions (consistent with $r = 0.63$ in Fig. 5), or reveal spurious reliance on weakly correlated features (temperature, pressure, humidity)? Regulatory acceptance of ML-based forecasting requires affirmative answers to these transparency questions, particularly given the proposed model's complexity relative to interpretable baselines like Random Forest.

Hybrid ensemble architecture. Combine the proposed neural framework with Random Forest through stacking or gating mechanisms, explicitly leveraging their complementary error profiles. Random Forest achieves the lowest baseline MAE (6.3 MW) through effective piecewise constant approximation of the cubic power curve at low wind speeds, while the proposed model dominates in RMSE (4.5 vs. 101.6 MW) and R^2 (0.99 vs. 0.83) through continuous nonlinear mapping. A regime-switching ensemble—selecting RF predictions during low-generation periods where its local approximation excels, and neural predictions during medium-to-high generation where temporal dynamics dominate—could theoretically achieve lower overall MAE than either standalone model while maintaining the neural framework's peak-tracking precision. This addresses the operational reality that no single architecture optimizes all metrics: the proposed model's 73% MAE reduction over Random Forest (1.7 vs. 6.3 MW) already represents near-optimal absolute performance, but ensemble methods may further refine error distribution across operational regimes.

Real-time adaptation protocols. Develop online learning mechanisms that update model weights as new meteorological observations arrive, rather than relying on static annual training. This is essential for climate adaptation: the current model trained on one-year Xinjiang data may degrade if subsequent years exhibit anomalous weather patterns (e.g., extreme heat events altering boundary layer stability, shifts in prevailing wind direction due to climate oscillations). Adaptive protocols should incorporate concept drift detection to trigger retraining when prediction residuals exhibit systematic bias shifts. Given the proposed model's low RMSE (4.5 MW), even minor degradation (e.g., 10% increase to 5.0 MW) may be operationally significant for grid ramp-rate constraints, necessitating proactive monitoring rather than reactive recalibration.

Supplementary Information

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Supplementary file 1.

Author contributions

Li hanyuan: Research methods, initial draft of the article, article illustrations, and article layout. Jie Wu: Article direction, organizational support, official draft of the article, article proofreading, Data collection, article verification, and text revision.

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Data availability
The data used to support the findings of this study are available from the authors upon request.

Informed consent
All investigators involved in the study were aware of the content to be published and agreed to publish it publicly.

Declarations

Ethics approval and consent to participate
This research does not involve any content of human ethics and morality, and all research content can be made public.

Consent for publication
Not applicable.

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